

PiTFiT: Payload-Level Bit-Flip Attacks on Stateless IoT Broadcasting Protocols

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Abstract—Stateless broadcasting protocols in IoT networks are vulnerable to payload-level bit-flipping attacks. We present two attack schemes: physical-layer corruption detected by CRC and network-layer corruption that bypasses CRC. Using a LoRa mesh testbed, we show how a compromised node silently propagates false data, revealing critical vulnerabilities and motivating lightweight integrity defenses.

Index Terms—Bit-Flipping Attack, CRC, IoT Security, LoRa, Low-Power Networks, Mesh Routing, Payload Corruption

I. INTRODUCTION

Low-power mesh networks have emerged as a foundational communication paradigm for mission-critical applications such as precision agriculture, disaster management, battlefield operations, patient localization, patient monitoring in healthcare [1] and smart city infrastructure [2]. These systems are particularly critical in GPS-denied environments such as tunnels, collapsed urban zones, and hostile military regions, where traditional localization methods and centralized infrastructures fail. In these scenarios, mesh topologies leveraging long-range, low-power technologies like LoRaWAN enable autonomous and resilient communication. With support from AI-based localization and routing frameworks, these networks maintain situational awareness and coordination in real time, even under infrastructure-deprived or adversarial conditions [3].

To address the energy and computational constraints inherent in such deployments, researchers have turned to stateless broadcast protocols like ECHO [4]. ECHO eliminates the need for route maintenance, acknowledgments, or topology discovery, enabling efficient packet propagation without explicit coordination. However, this control-free nature introduces significant security trade-offs: the protocol lacks higher-layer validation and semantic integrity checks, leaving these networks highly vulnerable to subtle, packet-level integrity violations.

Real-world hacker attacks have demonstrated how these vulnerabilities can be exploited to compromise mission-critical

systems. For example, the Stuxnet worm manipulated payload data in industrial control networks to sabotage nuclear centrifuges without detection [5]. GPS spoofing attacks on military drones and ships have silently altered geolocation data, resulting in misallocation of resources and operational failures [6]. In the IoT domain, bit-flipping and payload-tampering attacks on smart meters have enabled hackers to falsify energy consumption data [7]. These examples underscore the urgency of securing stateless IoT broadcast protocols against payload-level corruption in adversarial environments.

Among these threats, payload-level bit-flipping attacks represent a particularly stealthy and underexplored challenge. Bit-flipping involves altering one or more bits within a transmitted packet [8]. These corruptions may occur unintentionally due to noise, radiation, or hardware faults, or deliberately via adversarial manipulation such as software fault injection [9]. We classify these attacks into two distinct types based on their OSI layer of impact: (i) bit-level corruption at the physical layer, and (ii) character-level corruption at the network layer.

At the physical layer, bit flips typically trigger Cyclic Redundancy Check (CRC) failures, enabling radios to detect and discard corrupted packets. In contrast, network-layer corruption occurs post-CRC validation, where a compromised node modifies in-memory payload data such as GPS coordinates or sensor readings. Because CRC is not recomputed and stateless protocols like ECHO lack semantic integrity checks, these corrupted packets propagate silently, leading to false data being accepted as legitimate [10].

This paper presents the first real-world experimental analysis and direct comparison of bit-level and character-level bit-flipping attacks on stateless LPWAN mesh networks. Using a physical five-node testbed built with ESP32 microcontrollers, RFM95 LoRa radios, and GPS modules [11], we systematically injected randomized corruption at both the physical and network layers. Our empirical results reveal a critical insight: while bit-level corruption is often detected and filtered by MAC-layer CRC mechanisms, character-level corruption bypasses these defenses entirely, allowing falsified data to

silently propagate across the mesh.

The novelty of this work lies in exposing a previously overlooked class of vulnerabilities in stateless broadcast protocols and demonstrating, through real hardware experiments, how lightweight, application-aware integrity mechanisms are essential to protect energy-constrained IoT networks from stealthy adversarial threats.

Unlike prior studies that rely primarily on simulations or theoretical analyses, this work is distinguished by its deployment of a real hardware testbed to examine the security implications of payload-level corruption. Existing research on LoRaWAN security has focused predominantly on gateway-centric attacks such as jamming, replay, and eavesdropping [9], often overlooking threats arising from compromised relay nodes within the mesh. In contrast, our experiments demonstrate how even a single malicious node in a stateless broadcast topology can silently propagate falsified data across multi-hop paths, undermining network integrity and situational awareness. This empirical evidence highlights a critical blind spot in current LPWAN security frameworks and motivates the development of lightweight integrity verification techniques tailored to resource-constrained IoT environments.

The rest of this paper is organized as follows. Section II reviews recent work on bit-flipping attacks and resilience strategies in low-power mesh networks. Section III details our hardware testbed, attack setup, and performance metrics. Section IV presents empirical findings on the impact of physical- and network-layer bit flipping and analyzes the security implications. Finally, Section V concludes the paper and emphasizes the need for robust yet lightweight mitigation strategies.

II. BACKGROUND

A. Stateless Protocols and the ECHO Framework

The ECHO protocol [4] represents a class of stateless broadcasting frameworks tailored for energy-constrained wireless networks such as LoRa-based mesh topologies. By design, ECHO avoids the use of control packets, route discovery, and acknowledgments—optimizing scalability and minimizing power consumption [4]. However, this stateless nature introduces critical trade-offs: the protocol lacks context awareness, inter-packet consistency checks, and replay protection [4]. Once a packet passes CRC validation at the physical layer, it is accepted and rebroadcasted without inspecting the semantic integrity of its payload or headers.

This design choice, while ideal for energy efficiency, creates a dangerous blind spot for cyber-physical systems operating in contested or untrusted environments. Unlike traditional routing protocols that employ cryptographic validation or redundancy mechanisms, stateless protocols like ECHO assume well-formed and trustworthy input—making them inherently vulnerable to subtle, in-band data corruption.

B. Cybersecurity Gaps in Stateless LPWAN Meshes

Prior work has documented attacks on LoRaWAN systems including spoofing, replay, and jamming [9], [12]. However,

these typically target centralized gateways or rely on MAC-layer manipulation. In contrast, our focus is on payload-level bit-flipping—a class of attacks where internal nodes, especially compromised relays, inject silent corruption into the payload or application-level metadata [13]. These modifications may affect hop count, node ID, GPS data, or sensor readings [14]. Because ECHO does not re-compute CRC or perform in-network validation post-broadcast [10], such corrupted packets propagate unchallenged, undermining decision logic at the application layer.

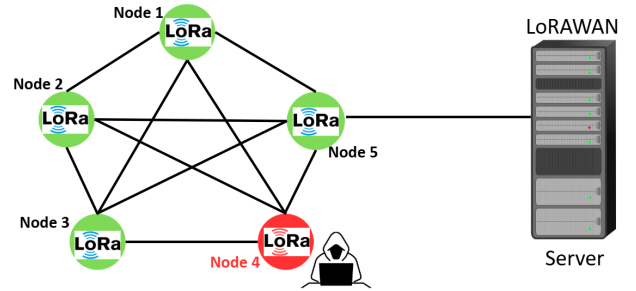


Fig. 1: Bit-flipping attack in a five-node LoRa mesh testbed. Node 4 is compromised and injects character-level and bit-level corruption into rebroadcasted packets.

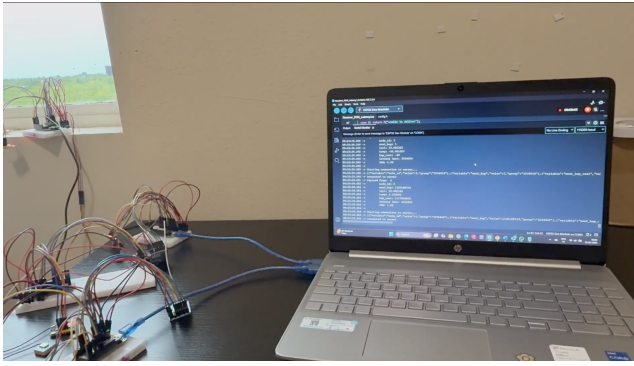
Bit-flipping can occur either at the physical layer—where CRC typically detects errors—or at the network layer, where corrupted payloads bypass detection entirely. This distinction is novel and underexplored in LPWAN literature, where most prior studies either simulate PHY faults or focus on higher-layer encryption schemes [15]–[18]. We provide a first-of-its-kind experimental breakdown of both bit-level and character-level corruption using a real-world mesh deployment of ESP32 nodes with RFM95 LoRa modules.

C. Novelty of Our Work

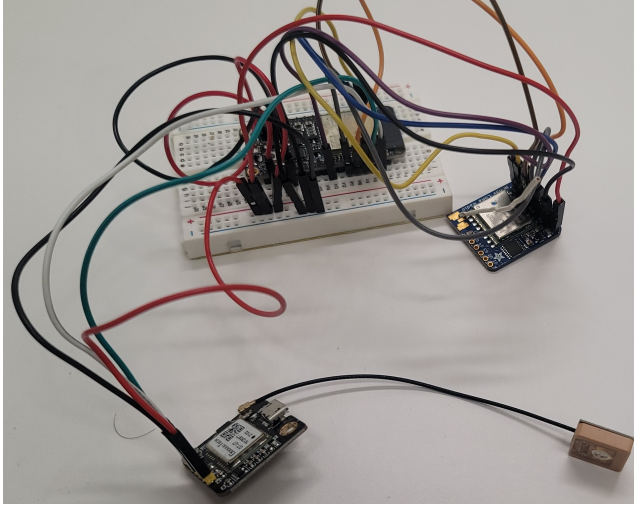
To the best of our knowledge, this paper is the first to:

- Experimentally distinguish physical-layer and network-layer bit-flipping attacks in stateless LoRa mesh networks.
- Demonstrate that character-level flips can evade CRC validation and corrupt mesh state silently.
- Quantify how bit-flipping propagates through stateless rebroadcasting, degrading metrics like PDR, latency, and semantic consistency.

Existing mitigation strategies, such as cryptographic payload authentication and deep learning-based anomaly detection, provide robust protection but are computationally intensive and unsuitable for resource-constrained IoT nodes [13], [9]. Prior studies often rely on simulations or focus solely on physical-layer corruption, overlooking stealthy character-level attacks that bypass CRC validation. In contrast, our work experimentally demonstrates these vulnerabilities on real hardware, highlighting limitations of existing defenses and motivating lightweight, application-aware integrity mechanisms. While



(a) Live data logging from ESP32 mesh nodes



(b) Node configuration using ESP32 + RFM95 + GPS

Fig. 2: Physical testbed used for bit-flipping attack evaluation. (a): Live data logging from ESP32 mesh nodes. (b): Node configuration using ESP32 + RFM95 + GPS. Five such nodes were deployed in a mesh with one attacker node performing random bit flips.

existing research has explored secure routing [2], geolocation [19], and anomaly detection [20], our work uniquely combines hardware-level attack simulation with protocol-level vulnerability analysis. The insights derived are not only critical for understanding LoRa's security posture but also for informing the design of future lightweight protocols that balance energy efficiency with cyber-resilience.

III. EXPERIMENTAL TESTBED CONFIGURATION

To rigorously evaluate the vulnerability of stateless IoT mesh networks to payload-level bit-flipping attacks, we constructed a real-world physical testbed that replicates typical LPWAN deployment conditions. The testbed as shown in the fig.2.(a) consists of five wireless sensor nodes configured in a multi-hop mesh topology. Each node is built using an ESP32 microcontroller integrated with an RFM95 LoRa transceiver operating at 915 MHz and a 6mm TinyGPS module to simulate positioning and movement data. Communication is governed by a lightweight, stateless broadcasting protocol inspired by

ECHO, which omits route discovery, acknowledgments, or integrity validation beyond MAC-layer CRC.

Out of the five nodes, one node (Node 4) is designated as a malicious actor. This node injected randomized bit flips into outgoing packets—targeting both the raw bitstream (bit-level corruption at the physical layer) and application-layer fields (character-level corruption at the network layer). The modifications have been performed either before transmission or after CRC validation, without recalculating the CRC hash. This deliberate omission ensured the silent propagation of corrupted packets, mimicking real-world vulnerabilities in stateless protocols.

The physical deployment and hardware arrangement of the testbed are illustrated in Fig. 2. The Fig. 2(a) shows the live serial logging of node behavior, while the Fig. 2(b) highlights the ESP32 + RFM95 + GPS stack used in each node.

The testbed was deployed indoors under controlled conditions for a duration of 60 minutes. During this period, metrics such as packet delivery ratio (PDR), bit error rate (BER), CRC failure rate, and end-to-end latency have been logged in real time. Key parameters of the testbed configuration are summarized in Table I.

TABLE I: Testbed Network Configuration Parameters

Parameter	Value
Number of Nodes	5 (1 attacker, 4 normal)
Microcontroller	ESP32 (240 MHz, 520 KB SRAM)
LoRa Chip	RFM95 (915 MHz)
GPS Module	TinyGPS (6mm)
Payload Size	32 bytes (fixed)
Spreading Factor (SF)	7
Bandwidth	125 kHz
Coding Rate	4/5
Transmission Power	14 dBm
Inter-Packet Interval	2 seconds
CRC Check	Enabled at MAC; not recomputed post-modification
Attack Type	Bit/char-level flipping (randomized)
Test Duration	60 minutes
Topology Type	Stateless mesh (ECHO-based)

This real-world setup allows us to capture both physical-layer and network-layer impacts of bit-flipping in stateless LPWAN protocols. By avoiding emulation and using physical devices, we ensure high-fidelity insights into packet loss dynamics, propagation of invalid data, and protocol vulnerabilities under realistic operating constraints.

IV. RESULTS AND ANALYSIS

To assess the impact of payload-level bit-flipping attacks in stateless LPWAN mesh networks, we conduct a series of controlled experiments on our five-node testbed. The attacker node (Node 4) randomly injects either bit-level or character-level corruption into the outgoing packet payloads. We focus on two critical dimensions of impact: geolocation integrity and performance degradation.

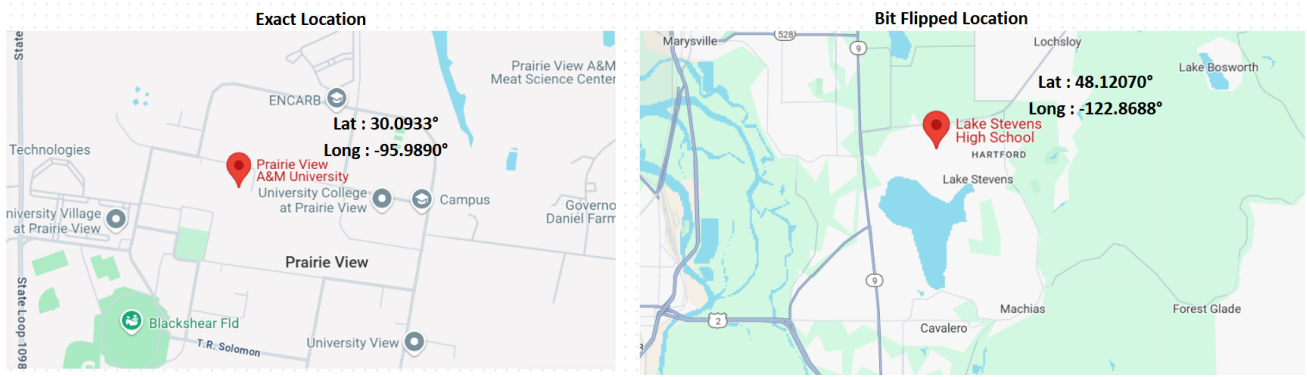


Fig. 3: Effect of character-level bit-flipping on GPS coordinates. The original position (left) is replaced with a location 2,700+ km away due to payload corruption.

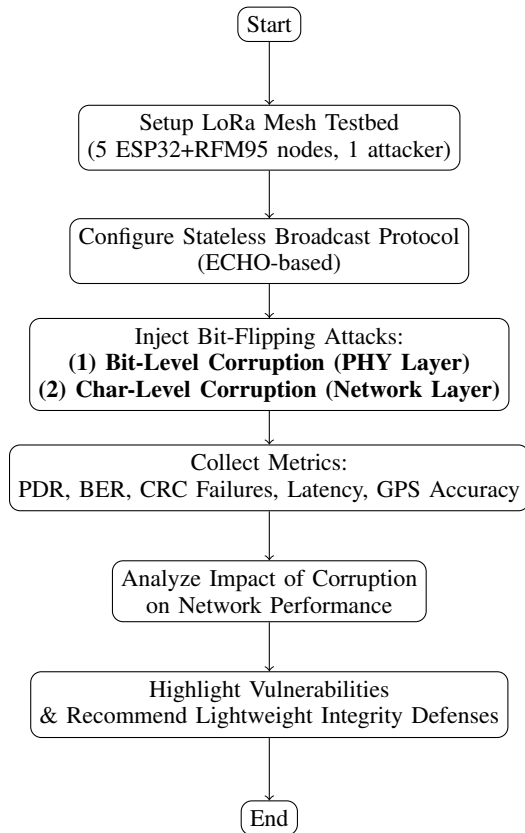


Fig. 4: Flowchart of Experimental Approach for Bit-Flipping Attack Analysis.

A. Impact on Geolocation Accuracy

Fig 3 demonstrates the real-world consequences of character-level bit flipping on GPS coordinate fields. The left-hand map shows the accurate location of the node at Prairie View A&M University in Texas (Lat: 30.0933°, Long: -95.9890°). However, after character-level corruption, the received coordinates placed the node near Lake Stevens High School in the State of Washington (Lat: 48.12070°, Long: -122.8688°).

This distortion—caused by flipping a few ASCII characters in the payload—bypassed CRC verification, and silently misled downstream applications. In critical scenarios such as disaster response or military tracking, such spoofed location data can result in misallocation of resources, failed coordination, or even physical risk.

B. Performance Metrics: Bit-Level vs. Char-Level Corruption

Fig. 5 illustrates the comparative impact of character-level and bit-level bit-flipping on four key performance metrics: Packet Delivery Ratio (PDR), CRC check success rate, Received Signal Strength Indicator (RSSI), and end-to-end latency.

The most notable observation lies in the contrast between character-level and bit-level corruption. Bit-level corruption—introduced at the physical layer—caused a substantial drop in PDR, from 0.84 to 0.40. This is expected, as hardware-level CRC mechanisms typically detect such corruption and discard the affected packets before they are forwarded. Correspondingly, the CRC success rate under bit-level attacks is also low, confirming that these packets are often eliminated at the PHY/MAC layer.

In contrast, character-level bit flips resulted in a deceptively high PDR of 84%. Although these packets contain corrupted payloads, the errors introduced after the CRC check has already validated the packet at the MAC layer. Consequently, the network continues to treat them as valid and forwards them without any integrity verification. This outcome highlights a significant vulnerability in stateless protocols like ECHO: their inability to inspect or reject packets with semantically corrupted data. As such, high PDR in this context does not imply trustworthy communication—it simply reflects unchecked propagation of potentially misleading or harmful information, such as altered GPS coordinates or falsified sensor readings.

RSSI values remain stable at approximately -65 dBm for both corruption types, indicating that bit-flipping affects data integrity but not signal strength. Latency, on the other hand, shows a slight increase from 140 ms to 150 ms under bit-level

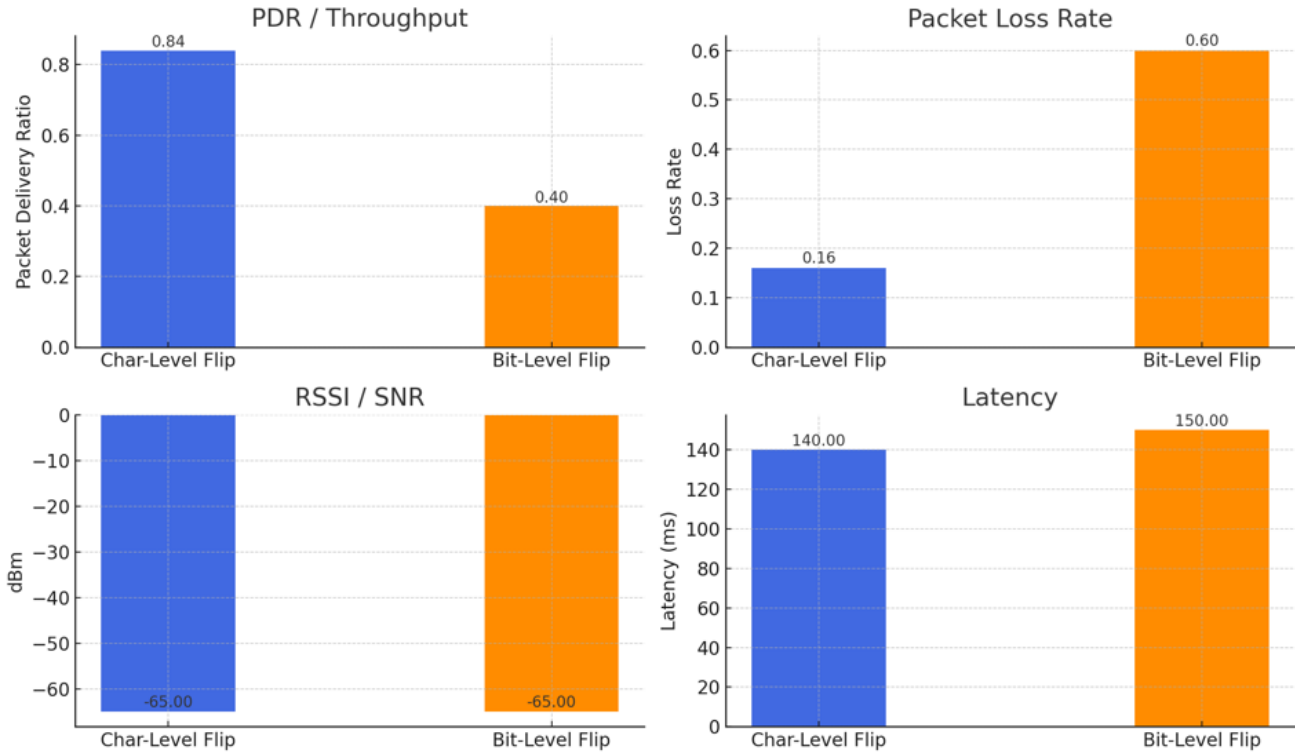


Fig. 5: Bit-Flipping Impact: Character-Level vs Bit-Level corruption on network metrics.

corruption, likely due to retransmissions or delays caused by CRC failure handling.

Overall, these findings validate our hypothesis: bit-level corruption is often caught and filtered early due to CRC mismatches, while character-level corruption is more insidious—passing silently through the network, corrupting application-layer logic, and undermining the integrity of mesh communications. This distinction is critical, especially in stateless LPWAN protocols that lack higher-layer validation. Together with our GPS spoofing analysis, the results emphasize the need for lightweight, application-aware integrity checks to defend against such silent threats in resource-constrained mesh environments.

V. CONCLUSION

This paper presents the first real-world experimental investigation into the vulnerability of stateless broadcasting protocols, such as ECHO, to payload-level bit-flipping attacks in low-power IoT mesh networks. By deploying a physical testbed comprising ESP32-based nodes equipped with LoRa and GPS modules, we systematically emulated two distinct attack schemes: bit-level corruption at the physical layer and character-level tampering at the network layer. Our findings reveal a critical security gap: while bit-level attacks are often detected and mitigated by MAC-layer CRC mechanisms, character-level attacks evade detection entirely, allowing corrupted payloads to propagate silently through the mesh.

This silent corruption has far-reaching implications for mission-critical applications such as battlefield communications, disaster response, and smart infrastructure, where false geolocation data or sensor readings can lead to misallocation of resources, operational failures, and even loss of life. Unlike prior work that relies on simulation or heavyweight cryptographic solutions, our study highlights the real-world feasibility of such attacks and underscores the urgency for lightweight, application-aware integrity mechanisms that can operate within the tight energy and computational budgets of IoT devices.

By exposing these vulnerabilities, we aim to inform the design of next-generation LPWAN protocols that balance energy efficiency with resilience against stealthy adversarial threats. The insights derived from this study are intended to serve as a foundation for developing scalable defense strategies to secure emerging IoT ecosystems deployed in contested or infrastructure-deprived environments.

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